

Name: _____

Date: _____

Mixed Polynomial Factoring

This prehistoric monument on Salisbury Plain in England consists of a massive ring of standing stones, each weighing around 25 tons.

Match your answer in the solution bank to find the letter for each blank.

Factor each expression completely:

Solution Bank

Factor completely:

(509) $x^2 + 7x + 10$

(142) $3x^2 + 15x$

(234) $x^2 - 36$

(612) $4x^2 - 12x$

(341) $x^2 - 49$

(815) $x^2 + 6x + 8$

(788) $x^2 + 8x + 15$

Solution

Letter

$(x - 6)^2$ [P]

$(x - 7)(x + 7)$ [H]

$3x(x + 5)$ [S]

$(x - 6)(x + 6)$ [O]

$4x(x - 3)$ [N]

$3x(x + 15)$ [K]

$(x + 2)(x + 4)$ [T]

$(x + 1)(x + 10)$ [W]

$(x + 2)(x + 5)$ [E]

$(x + 3)(x + 5)$ [G]

142

815

234

612

509

341

509

612

788

509



Name: _____

Date: _____

Mixed Polynomial Factoring

These massive ancient geoglyphs in the Peruvian desert depict animals and plants, and are best viewed from the air.

Match your answer in the solution bank to find the letter for each blank.

Factor each expression completely:

Solution Bank

Factor completely:

(405) $2x^2 - 50$

(166) $3x^2 - 15x + 18$

(918) $5x^2 - 20$

(892) $3x^2 - x - 2$

(239) $2x^2 - 5x - 3$

(782) $4x^2 + 12x - 40$

(314) $2x^2 + 16x + 30$

(527) $2x^2 + 7x + 3$

Solution

Letter

$4(x + 5)(x - 2)$

[L]

$(2x + 3)(x + 1)$

[W]

$(3x + 2)(x - 1)$

[I]

$2(x - 5)(x + 5)$

[N]

$3(x - 2)(x - 3)$

[A]

$(2x + 1)(x - 3)$

[S]

$5(x - 2)(x + 2)$

[C]

$2(x^2 - 25)$

[M]

$3(x - 1)(x - 6)$

[P]

$2(x + 3)(x + 5)$

[E]

$(2x + 1)(x + 3)$

[Z]

405

166

527

918

166

782

892

405

314

239



Name: _____

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Mixed Polynomial Factoring

This remote location in the Pacific Ocean is famous for its nearly 1,000 monumental stone statues, called moai, created by the early Rapa Nui people.

Match your answer in the solution bank to find the letter for each blank.

Factor each expression completely:

Solution Bank

Factor completely:

(157) $25x^2 - 64y^2$

(349) $9x^2 - 100y^2$

(571) $9x^2 - 12x + 4$

(923) $4x^2 - 4x - 15$

(834) $-6x^2 - 2x + 4$

(718) $144x^2 - 121$

(481) $-6x^2 + 15x + 9$

(268) $6x^2 + 7x - 5$

(605) $-8x^2 + 12x + 20$

Solution

Letter

$(12x - 11)(12x + 11)$

[L]

$(2x - 1)(3x + 5)$

[I]

$(2x + 3)(2x - 5)$

[S]

$(3x - 10y)(3x + 10y)$

[T]

$-3(2x - 1)(x + 3)$

[B]

$(3x - 2)^2$

[D]

$(2x + 1)(3x - 5)$

[M]

$-2(3x - 2)(x + 1)$

[R]

$(5x - 8y)(5x + 8y)$

[A]

$-3(2x + 1)(x - 3)$

[E]

$(5x - 8y)^2$

[F]

$-4(2x - 5)(x + 1)$

[N]

481

157

923

349

481

834

268

923

718

157

605

571



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STONEHENGE

Factor each expression completely:

Solution Bank

Factor completely:

(509) $x^2 + 7x + 10$

$(x + 2)(x + 5)$ 509 [E]

(142) $3x^2 + 15x$

$3x(x + 5)$ 142 [S]

(234) $x^2 - 36$

$(x - 6)(x + 6)$ 234 [O]

(612) $4x^2 - 12x$

$4x(x - 3)$ 612 [N]

(341) $x^2 - 49$

$(x - 7)(x + 7)$ 341 [H]

(815) $x^2 + 6x + 8$

$(x + 2)(x + 4)$ 815 [T]

(788) $x^2 + 8x + 15$

$(x + 3)(x + 5)$ 788 [G]

Solution

Letter

$(x - 6)^2$

[P]

$(x - 7)(x + 7)$

[H]

$3x(x + 5)$

[S]

$(x - 6)(x + 6)$

[O]

$4x(x - 3)$

[N]

$3x(x + 15)$

[K]

$(x + 2)(x + 4)$

[T]

$(x + 1)(x + 10)$

[W]

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Date: _____

Mixed Polynomial Factoring

These massive ancient geoglyphs in the Peruvian desert depict animals and plants, and are best viewed from the air.

NAZCA LINES

Factor each expression completely:

Solution Bank

Factor completely:

(405) $2x^2 - 50$
 $2(x - 5)(x + 5)$ 405 [N]

(166) $3x^2 - 15x + 18$
 $3(x - 2)(x - 3)$ 166 [A]

(918) $5x^2 - 20$
 $5(x - 2)(x + 2)$ 918 [C]

(892) $3x^2 - x - 2$
 $(3x + 2)(x - 1)$ 892 [I]

(239) $2x^2 - 5x - 3$
 $(2x + 1)(x - 3)$ 239 [S]

(782) $4x^2 + 12x - 40$
 $4(x + 5)(x - 2)$ 782 [L]

(314) $2x^2 + 16x + 30$
 $2(x + 3)(x + 5)$ 314 [E]

(527) $2x^2 + 7x + 3$
 $(2x + 1)(x + 3)$ 527 [Z]

Solution

Letter

$4(x + 5)(x - 2)$ [L]

$(2x + 3)(x + 1)$ [W]

$(3x + 2)(x - 1)$ [I]

$2(x - 5)(x + 5)$ [N]

$3(x - 2)(x - 3)$ [A]

$(2x + 1)(x - 3)$ [S]

$5(x - 2)(x + 2)$ [C]

$2(x^2 - 25)$ [M]

$3(x - 1)(x - 6)$ [P]

$2(x + 3)(x + 5)$ [E]

$(2x + 1)(x + 3)$ [Z]

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Mixed Polynomial Factoring

This remote location in the Pacific Ocean is famous for its nearly 1,000 monumental stone statues, called moai, created by the early Rapa Nui people.

EASTER ISLAND

Factor each expression completely:

Solution Bank

Factor completely:

(157) $25x^2 - 64y^2$

$(5x - 8y)(5x + 8y)$ 157 [A]

(349) $9x^2 - 100y^2$

$(3x - 10y)(3x + 10y)$ 349 [T]

(571) $9x^2 - 12x + 4$

$(3x - 2)^2$ 571 [D]

(923) $4x^2 - 4x - 15$

$(2x + 3)(2x - 5)$ 923 [S]

(834) $-6x^2 - 2x + 4$

$-2(3x - 2)(x + 1)$ 834 [R]

(718) $144x^2 - 121$

$(12x - 11)(12x + 11)$ 718 [L]

(481) $-6x^2 + 15x + 9$

$-3(2x + 1)(x - 3)$ 481 [E]

(268) $6x^2 + 7x - 5$

$(2x - 1)(3x + 5)$ 268 [I]

(605) $-8x^2 + 12x + 20$

$-4(2x - 5)(x + 1)$ 605 [N]

Solution

Letter

$(12x - 11)(12x + 11)$

[L]

$(2x - 1)(3x + 5)$

[I]

$(2x + 3)(2x - 5)$

[S]

$(3x - 10y)(3x + 10y)$

[T]

$-3(2x - 1)(x + 3)$

[B]

$(3x - 2)^2$

[D]

$(2x + 1)(3x - 5)$

[M]

$-2(3x - 2)(x + 1)$

[R]

$(5x - 8y)(5x + 8y)$

[A]

$-3(2x + 1)(x - 3)$

[E]

$(5x - 8y)^2$

[F]

$-4(2x - 5)(x + 1)$

[N]

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